

Take-Home: Comm-Log Send Reconciliation

Time Allotted: Assignment submission will close on Monday

Background

Finance tracks a metric called `target_base`: the total number of qualifying sends for a merchant's campaigns over a period. For **merchant 501, October 2026, across all Diwali campaigns**, Finance says: **The true `target_base` is 22.**

You've been asked to reproduce this number from the raw data and explain any gap between it and a straightforward query.

Data

See README.md for the full schema and data dictionary. Raw data lives in `data/comm_log.db` (SQLite) or the equivalent CSVs in `data/`.

How to use data files

SQLite (`data/comm_log.db`) — recommended, since you can just write SQL directly:

```
Shell
sqlite3 "data/comm_log.db"
.tables                # campaign, communication_log
.schema campaign
.schema communication_log
```

Or run a query straight from the shell:

```
Shell
sqlite3 "data/comm_log.db" "SELECT COUNT(*) FROM
communication_log;"
```

CSVs (`data/campaign.csv`, `data/communication_log.csv`) — same data, in case you'd rather use pandas/Excel/whatever:

Python

```
import pandas as pd
campaign = pd.read_csv("data/campaign.csv")
comm_log = pd.read_csv("data/communication_log.csv")
```

What to submit

1. **A reconciliation bridge** (a simple table is fine): starting from the most naive/obvious query you could write against this data, list every adjustment you made to get to the final number, **in the order you discovered you needed it**, with a one-line reason for each.

Example shape:

Step	Description	Result	Reason
0	Naive count	X	(starting point)
1	...	Y	...
...	...	...	...
final		22	

2. **The SQL query (or queries)** you used to compute the final, correct number — something we can run ourselves against data/comm_log.db.
3. A short paragraph (a few sentences) on anything in the data that surprised you, even if it didn't end up changing your final number.

What we're evaluating

Not just whether you land on 22 — a bridge that mentions no real adjustments but somehow arrives at the right number is a red flag, not a pass. We want to see how you investigated: what you tried first, what didn't match, what you checked next, and why.

Here is the drive folder with all the supporting material:

 Data Analyst Assignment